

Database

→ Definitions

→ FIS (File Information System)

→ Keys

Manual File System

→ Pen & paper

Data Storage

↳ Record data on paper

↳ Paper, folder, cabinets.

Data Retrieval

↳ Locate paper from cabinets, folders

↳ Retrieve from paper

Disadvantage

↳ Wastage of time

↳ Wastage of resources

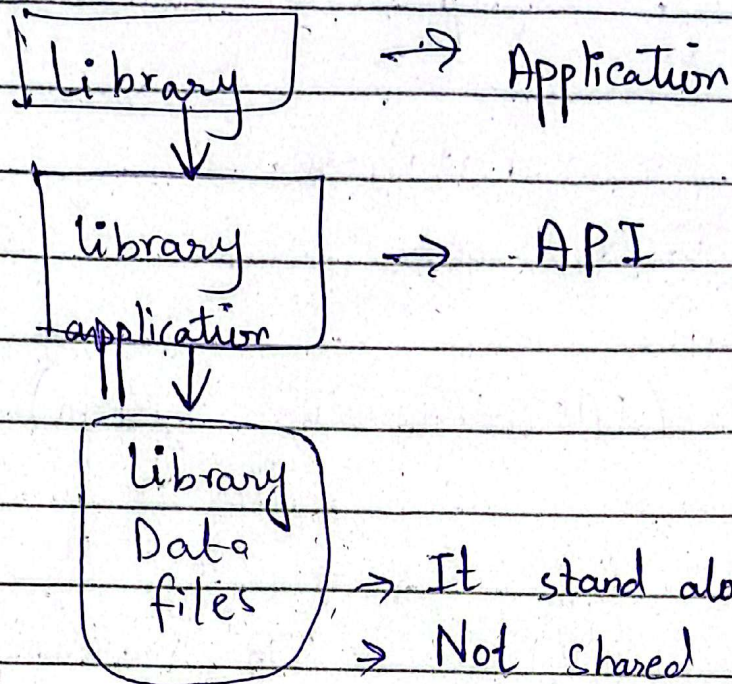
↳ More information to handle

Computerized File Management System

↳ End user (multiple users)

↳ No of users are infinity

→ Production of reports.



Disadvantage

- Duplication of Data
- Vulnerable to inconsistency

Database System

- Users
- DBMS
- Hardware (OS)

Database Management System

- DA → (Data administrator)
- DBA
- Database Administrator

Advantages of Database

Persistent

↳ Last for long time

Shared

↳ Can be accessed by multiple programs

Interrelated

↳ Data combined to form new data



(should be logically related)

Data & Information

→ Records

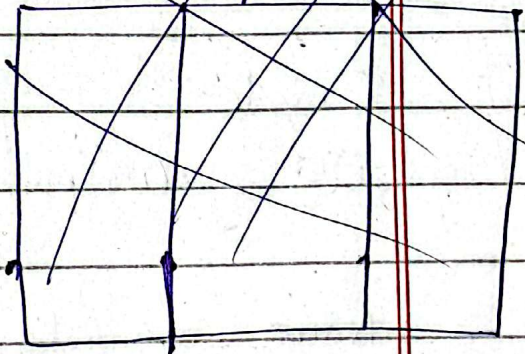
→ Instances

↓
cell k andar jo field
k against values hon
gi

(1 field k against same
value ho gi)

(1 record k against same
value ni ho gi)

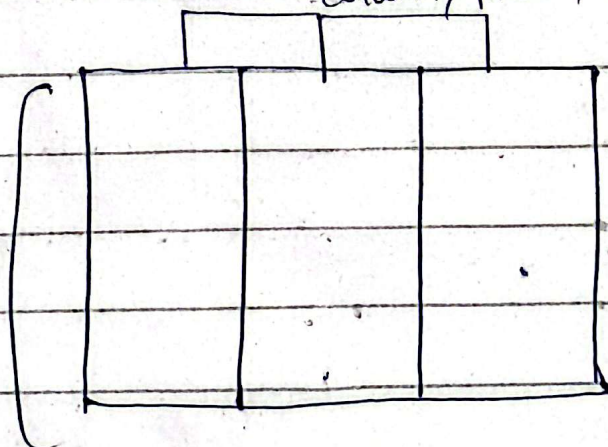
~~Table / Entity~~



~~Table / Entity~~

column / field / attribute

Row
Record
Tuple



Student (Entity)

	Name	Reg	Class	Marks	Stale	5 (Attributes)
Instances (Record)	-	-	-	-	-	
	-	-	-	-	-	
	-	-	-	-	-	
	-	-	-	-	-	
	-	-	-	-	-	

character
VARCHAR
variable

(LAB) ↓

→ Overall data (Schema)

↓
Entity (e.g sales)

e.g
ali ahmed = VARCHAR / CHAR
int [INT (5)]

Oracle
→ ~~sys~~ sys(123) password
float [Number ()]

⇒ Name of field is declared by datatype

LAB (Main Heading size 16)

SQL

(Times New Roman) + Bold

(^{sub heading}~~writing~~ size 14)

(writing 12)

→ Structure (Rows & columns)

→ fetch (from original database)

SQL (Structured Query Language)

→ Storing

→ Manipulating (Changing)

→ Retrieving

(Table / Relations)

Multiple relations (RDBMS) →

Relational Database Management system

DBMS (software hota hai)

MySQL

MS Access (Database RDBMS)

→ SQL Commands

DDL (Data Definition Language)

DML

DCL

DQL

DDL

→ Data ko define krna

CREATE TABLE Std_info (
Reg# VARCHAR(15) ^{koi bhi ye e.g hai}

Reg# VARCHAR2 (20).

std Name VARCHAR2 (20),

DOB DATE,

~~E-mail VARCHAR2 (30);~~

BSAT ~~IV~~ (Reg #, Name, DOB, E-mail, Phone no, CNIC)

~~CREATE TABLE std m~~

CREATE TABLE BSAI-IV (

Reg # VARCHAR2 (10),

Name VARCHAR2 (30),

DOB DATE, 

E-mail VARCHAR2 (50),

Phone # INT ~~XXXX~~,

CNIC VARCHAR(2);

BSAT - IV

Reg #	Name	DOB	E-mail	Phone #	CNIC

Diagram

→ Rectangular Box (Entity)

→ Scheme

↳ Information abt. structure of data is stored.

→ Database Applications

↳ API main kisi b language main program likh k task perform krskty

→ DBMS (Oracle main HR jo sab kuch kr rha)

↳ Management of data (petly user ko manage kryga then data ko)

↳ User of database

eg → LMS is DBMS

Advantages of DBMS

→ Data sharing

→ Controlled Redundancy (Duplication of data is called)

→ Better Data integrity

→ Data Consistency

→ Better data security

→ Better data accessibility

→ Improve data standards

Disadvantages

→ Increase cost

↳ Hardware cost

↳ Personal cost

↳ Adoption cost

→ Complexity

→ The great impact of failure.

→ More difficult recovery

V. IMP

Functionality of DBMS

→ Data storage & retrieval. (Delete main agar puri kr dain to record delete hogi)

→ Data Access language (LMS eg)

→ APIs, → Database communication Interface (communication of 2 components)

→ Data Integrity Management (follow rules)

→ Multi-user access control (concurrency)

→ Security Management (hashing, passwords etc.)

→ Backup & Recovery Management (Deleted files can be recovered)
(If no backup then no recovery)

(Any website jiska login ho wo database hai).

Data As Resource

Resource

↳ Anything that is beneficial to an organization & requires cost.

Q Is Data a resource?

YES

Levels of Data

→ Real world entities (Real entities)

→ Metadata (Schema)

→ Data occurrence (Data itself) (Meta data jis sy belong krta wo real world entity hgi)

Real world entities ~~Sales~~ Company XYZ
↳ ~~Sales Company XYZ~~

Meta data ~~Company Name~~ Sales
↳ ~~Company Name Sales~~ (Hpt) ♥

Data occurrence

↳ Records

Data Base Users

→ Application Programmers

→ End users

↳ Naive Users (who have knowledge but doesn't "technical knowledge")

↳ Sophisticated user (who have technical knowledge & programming knowledge)

→ Database Administration (DBA)

↳ Who have control over Database
e.g. (HR)

Functions of DBA

Schema definition

Granting data Access

length
→
precise
→
scale
→

variable ki length (4000 by default)
solid numeric / integer value (not decimal)
salary \$ eg. (after point)

45000 . 625

precise (5) scale (3)

number (5, 3)

varchar (50) 1 to 50 (0 is null)

1 table k andar multiple primary
keys ho sakti hain pr select
1 ho gi.

primary key can only be one
unique
not null

Theory Database

3-level

External level

User 1
view₁

User 2
view₂

User n
view_n

Conceptual level

Conceptual
Schema

Internal level

Internal
Schemas

Physical Data
Organization

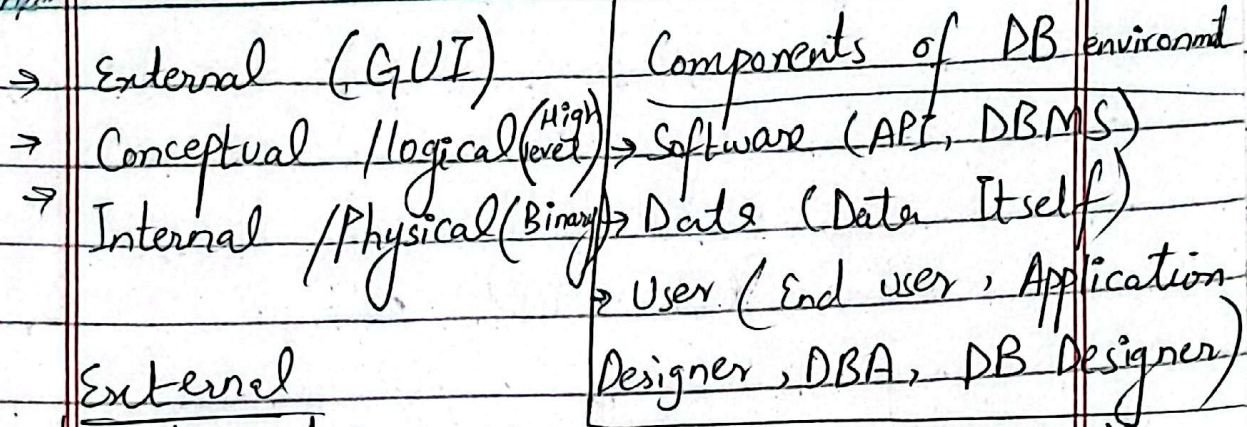
Data
base

e.g. ① Username ② Name ③ password ④ Confirm
 Password

then after confirmation will show

① & ③ checked by ② & ④

Three level Architecture

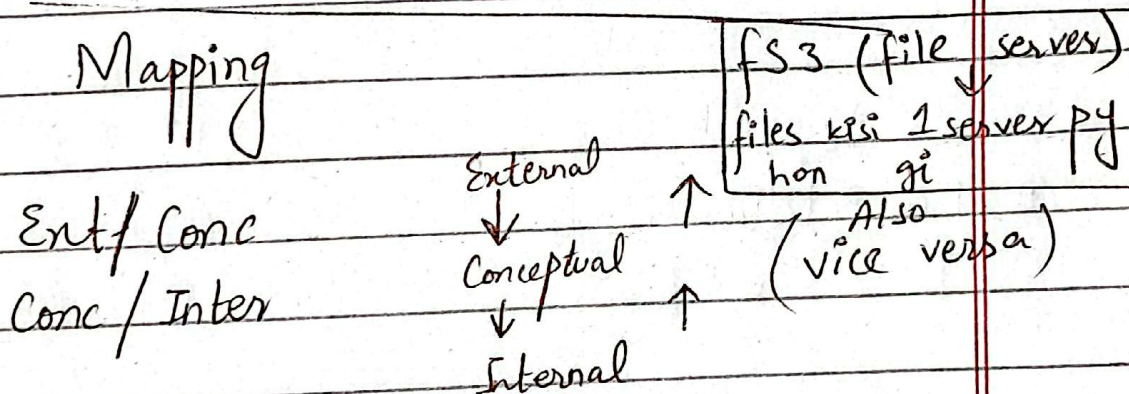


External
↳ Understood by user & DBA

Conceptual
↳ (Understood by ~~DBA~~ DBA)

Internal (Row header, Binary header, RH, BH)
↳ (Understood by Machine)
↳ Converted into Binary [or 1]

Mapping



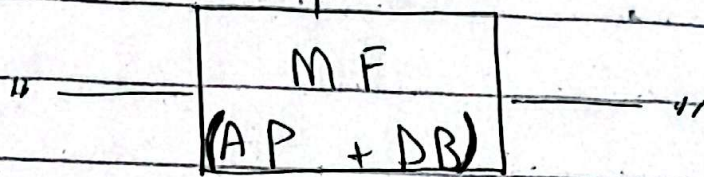
→ Change in conceptual doesn't affect external.

Physical Data Independence

→ Change in internal doesn't affect conceptual

Centralized Architecture

Display Terminal (dumb)



2 components

→ Main Frame

→ Dumb Terminals / Display Terminals

Client - Server (3 types hoti)

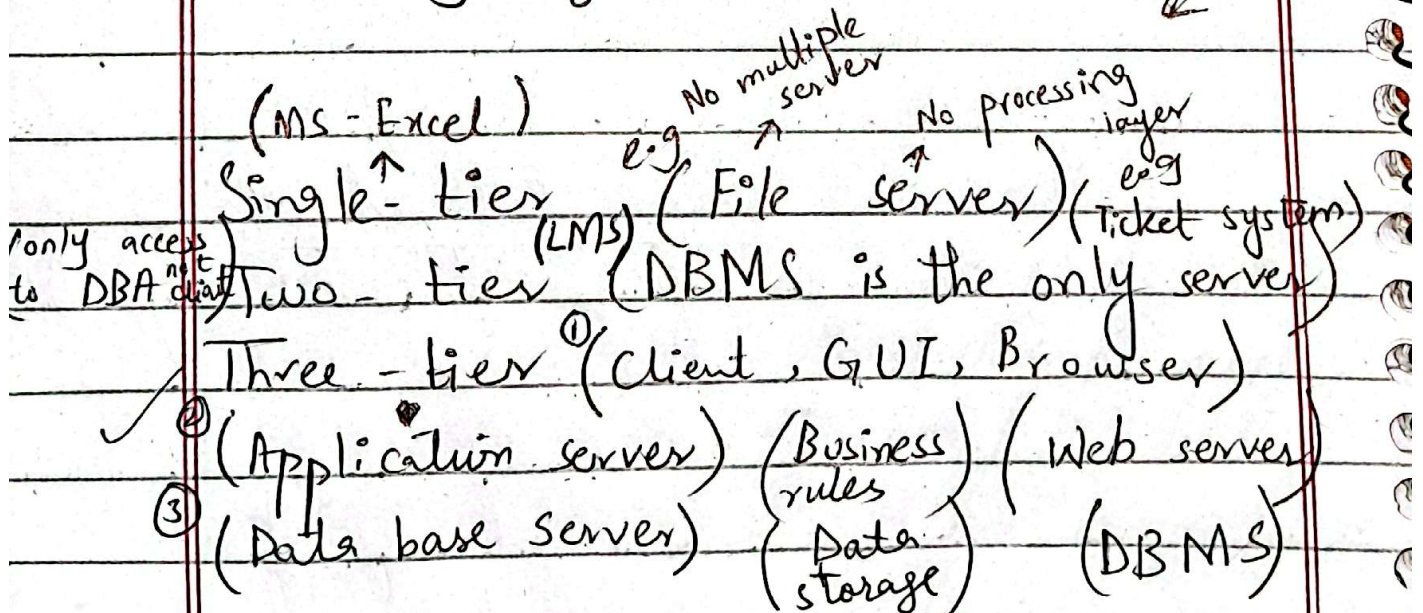
→ A main server

→ Multiple clients are connected to it through APIs.

3 layer

① Presentation layer

② Processing logic



File ^{server} ~~storage~~

- File storage
- Record locking
- Access to hard disk to client
- Busy or not

Middle ware

- ↳ Software allows application to connect with other
- ↳ APIs (third party)

Types

① Remote procedure calls (RPC)

- ↳ client make calls to procedures
- ↳ Asynchronous, synchronous

② Msg oriented Middleware (MOM)

- ↳ Asynchronous calls b/w client via msgs.

③ Publish / Subscribe

- ↳ Push technology (Server send info to client when available)

④ Object Request Broker (ORB)

- ↳ object oriented management of communication b/w clients & servers. e.g. (Banking card system)

⑤ SQL oriented data Access

↳ Middleware b/w applications & database servers-

Data base Middleware

① ODBC (open data base connectivity)

② JDBC (Java database connectivity)

DB Environment

- Single user

- Multi users

DB system Deployment

→ Setting up DB system in client / server architecture

→ 3 ways

① All on same computer (Excel)

② Multiple client 1 server (LMS)

③ Multiple client Multiple server (ATM)

Database System Components

- Software
- DB Designer
- DB Administrator
- User

Preliminary
Requirement Analysis

Tools

DFD (Data Flow Diagrams)

- ↳ Content diagrams
- ↳ level 0, 1 diagrams

Data store

↳ output

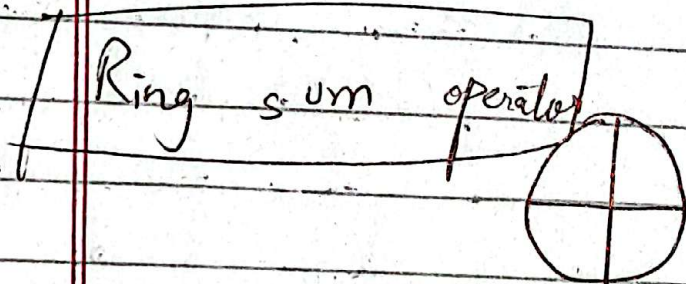
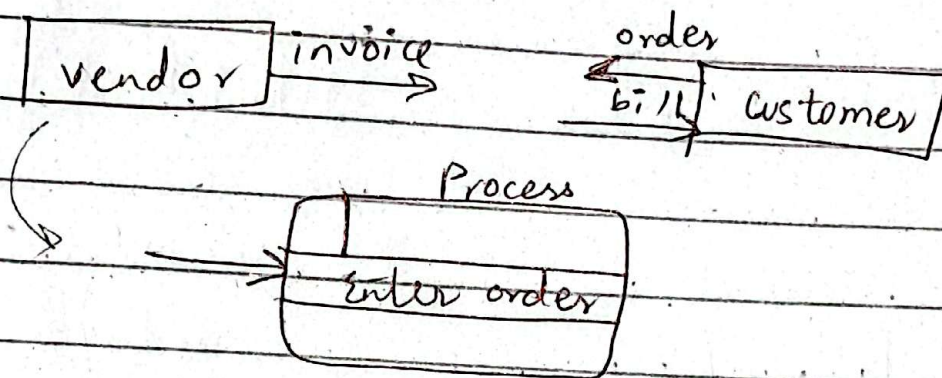
label ←  } Rectangle (shows entity/table)

Processing =  , 

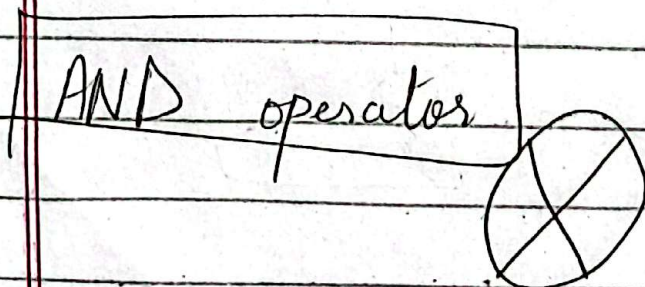
DFD Symbols

- ① External entity
- ② Process
- ③ Data Store

→ Entities kabhi bhi direct communicate ni kr sakti



- 2 data flows
- 1 is followed



- 2 data flows
- Both are followed

Types of DFD

- ① Context diagram (1 process multiple entities)
- ② level 0 "
- ③ level 1 & 2 "

ERD (user based scenario)

Entity :- (Physical)

↳ Anything that exist in a system is called entity

e.g. ↳ Person, place, object, event

Entity types

Person

Faculty, student, employees etc.

Place

Japan, India etc.

objects

Board, switch etc.

Event

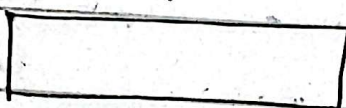
class (Any room, time etc.)

Entity set

↳ No. of instances when combined form entity set.

Entity Representation

↳ Entity Name will always be single word and ^{whole} word capital.



Rectangular box

e.g

STUDENT

FACULTY

⇒ Entity have attributes (characteristics / Properties or characteristics of an entity is called Attributes).

Data Modeling

↳ Process of creating a representation of data.

ER diagram

- ① Entity type
- ② Attributes
- ③ Relationship

Activity

Entity type :- ① Student
② Subject

STUDENT

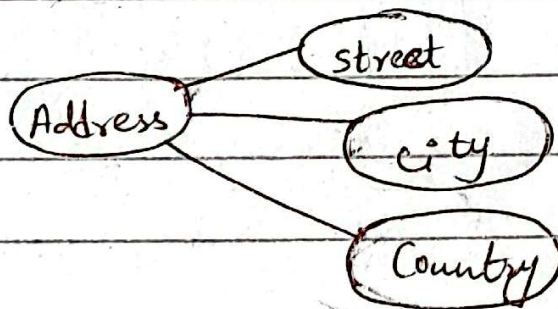
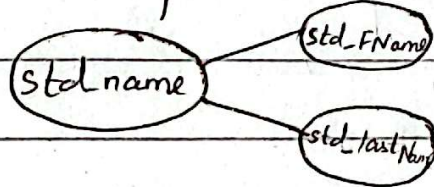
SUBJECT

Entity { ID
NAME
CONTACT No.
set (aka homo)

Attributes

- ① Simple / Single (Having only 1 attribute) ^{e.g ID}
- ② Composite (can be further divided)

e.g

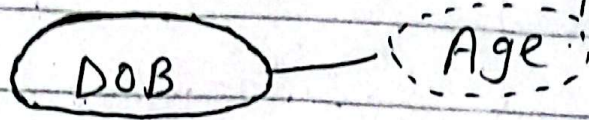


- ③ Multivalued (Have multiple values)

(std-Phone) (std-email) (std-skills)

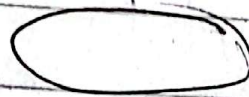
⇒ Address can also be multivalued e.g
(Permanent & Temporary)

④ Derived (Deriving ~~one~~ ^{other} value from given)



⑤ Stored (All values are already stored)

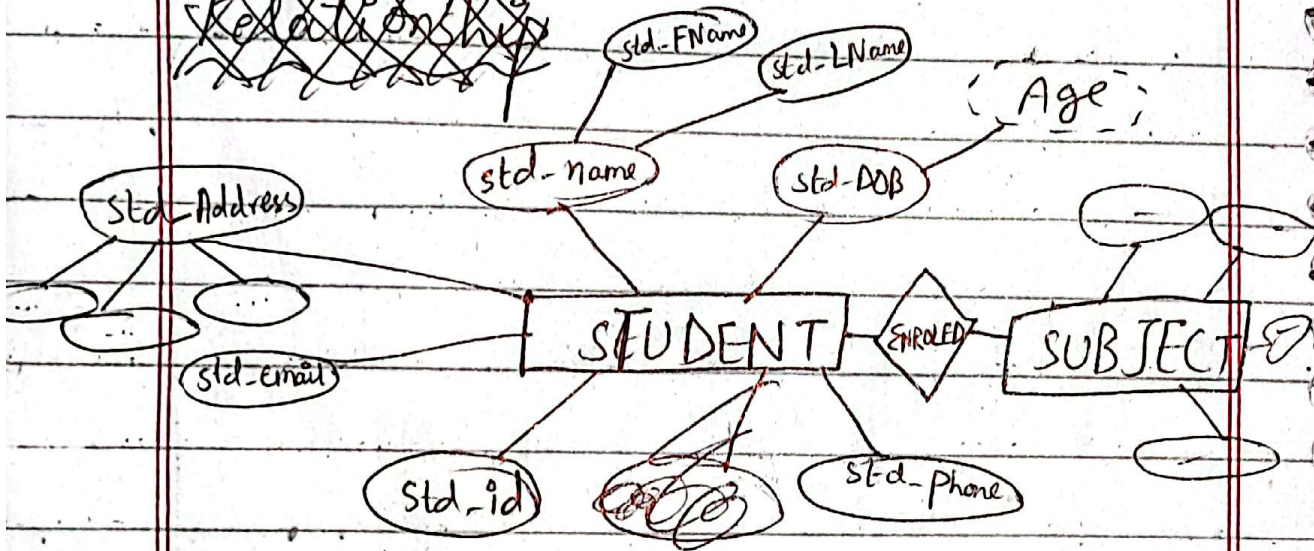
⇒ Representation of attributes



↓
oval

Relationship

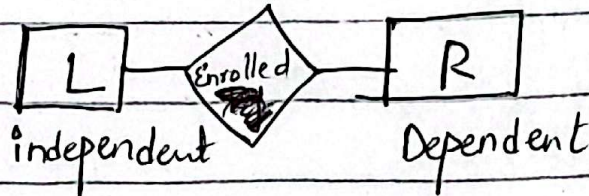
~~Relationship~~



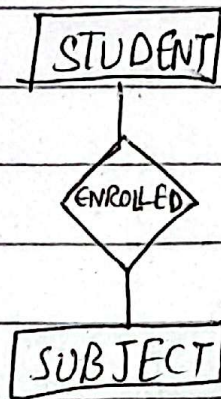
Rules to make ERD

① Left to right

(agar change krain to "ENROLLED BY")

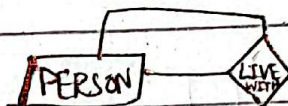


② Top to bottom



Types of Relationship

① Unary (1)



② Binary (2)



③ Ternary (3)



(To weakened 2 strong relationships we have to make 3rd)

⇒ Total no. of entities ^{types} is called degree.

⇒ There can be multiple key attributes but only one is selected as primary key.

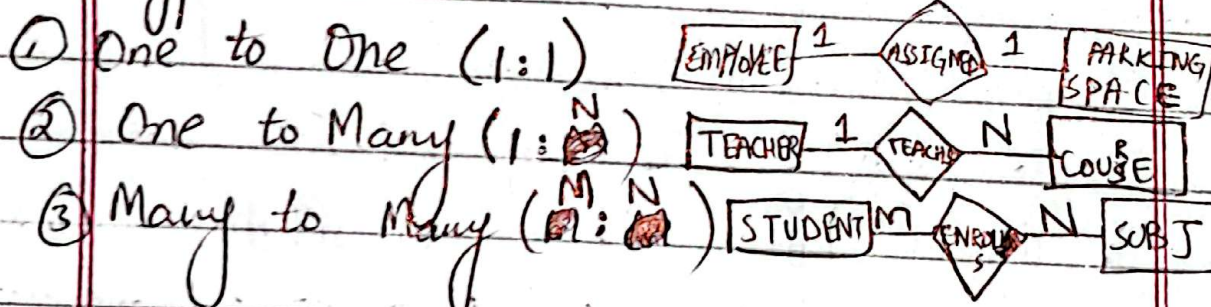
⇒ Two attributes combined together can't make composite attribute. (one must be non-key)

DBMS (Physical construction)
DB Designer (Map updation)
User

Cardinality (Atleast, utmost, exactly)

↳ No. of entities in a relationship

Types.



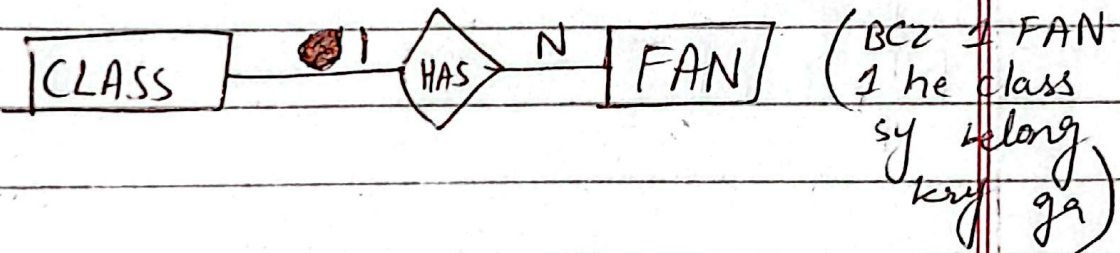
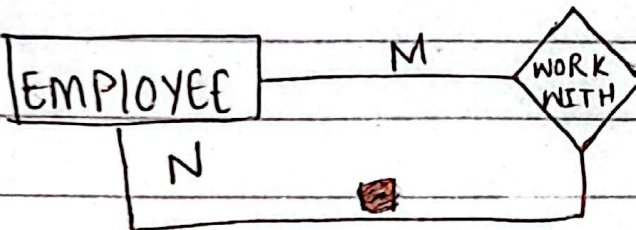
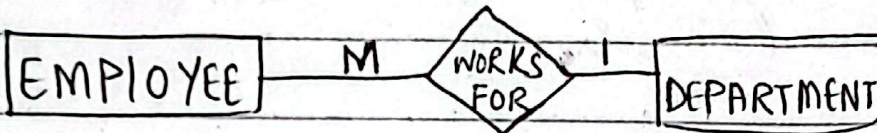
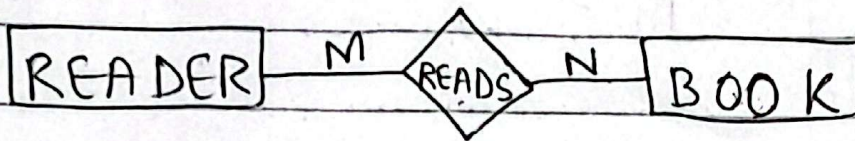
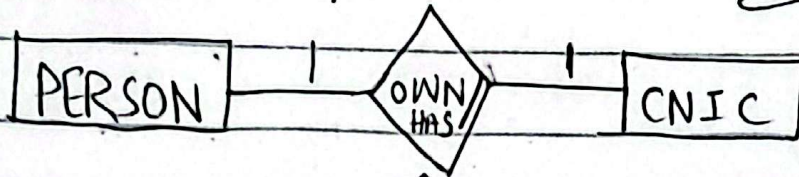
⇒ (Many to one (N:1))

Modality

Total (Must/Compulsory) ^(Double line) e.g. 

Partial (Maybe) ^(- single line)

Examples of Relationships

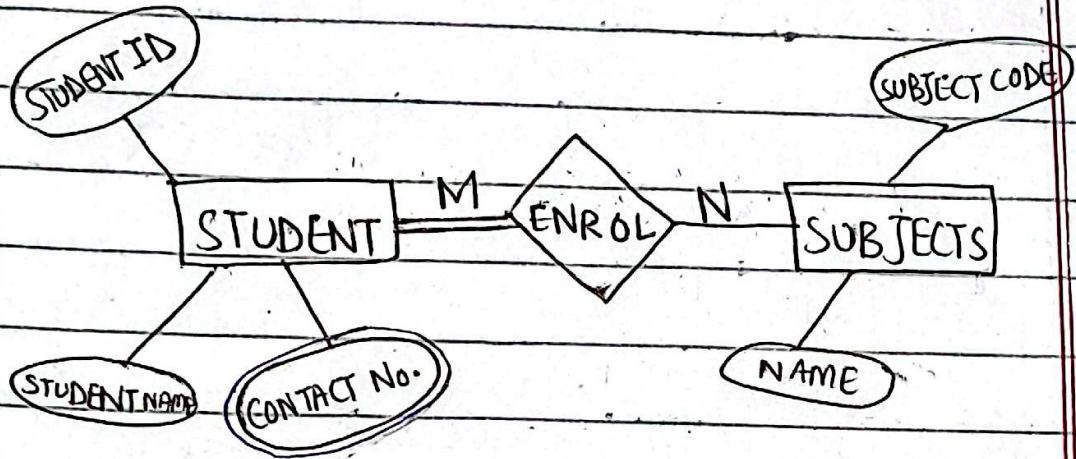
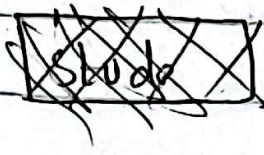


⇒ ERD is a user-based scenario.

(ID, NAME, CONTACT) (CODE, NAME)

Entity = STUDENT, SUBJECT

①



②

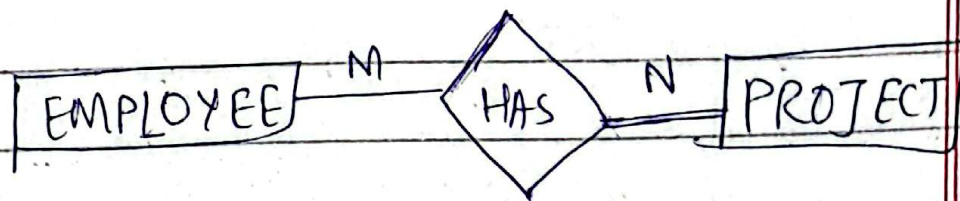
Entity &
Employee
Department



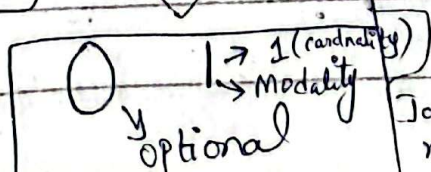
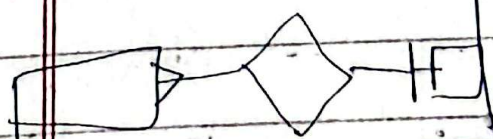
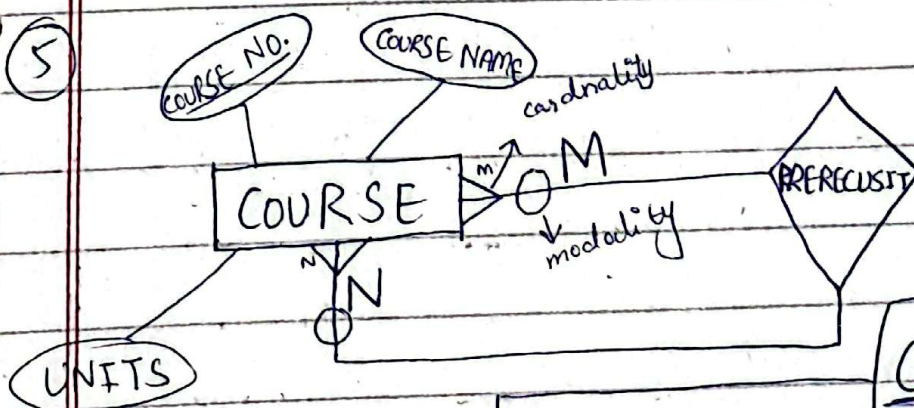
③ Entity
PATIENT
HISTORY



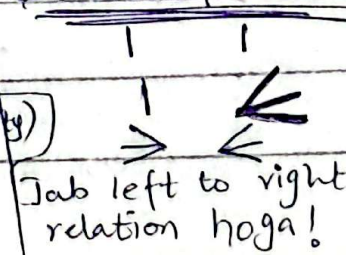
④ Entity ~~Company~~, Employees, Project



⑤



Crow's Foot



Chen - One

1:1
1:N
M:N

Entity

①

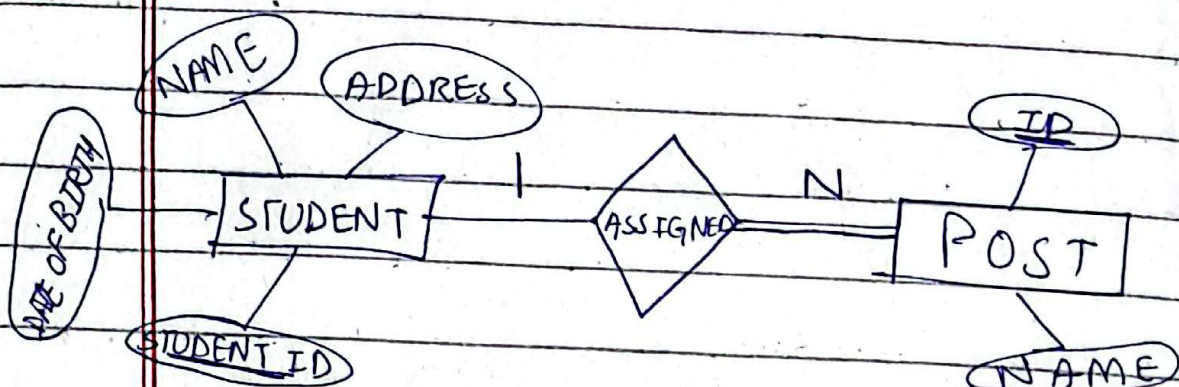
Student

↳ name, address, Date of Birth, Student ID

②

Post

↳ ID, Name



EERD

→ More enhanced or complicated ERD ko easy banany k liye usy more explain kr deta.

↳ subclass

↳ super class

super class k attributes subclass membership ki bhi hon gy but membership hona zaroori hai.

→ Inheritance (Super class ka hona lazmi hai)

subclass ka kabhi bhi seperate attribute ya ~~relationship~~ relationship ni banana.

Super Class

↳ Jis sy multiple types nikal ri hon. (Sub class)

④ 4 things

① Membership

② Attribute Inheritance

③

④ Local Attribute & Relationship

Specialization

↳ Top to bottom process.

start with

↳ Super class (honi chahiye)

↳ Identify karo

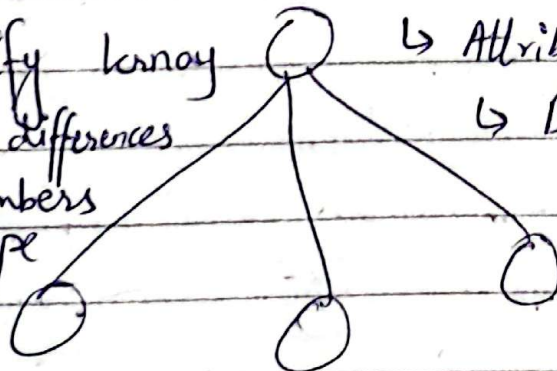
hairr. differences

how members

of entity type

↳ Attributes & relationship

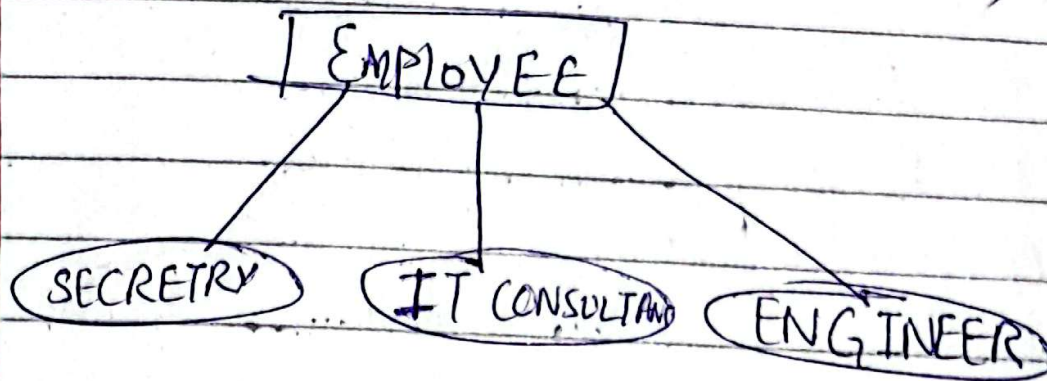
↳ Describe subclass



Generalisation

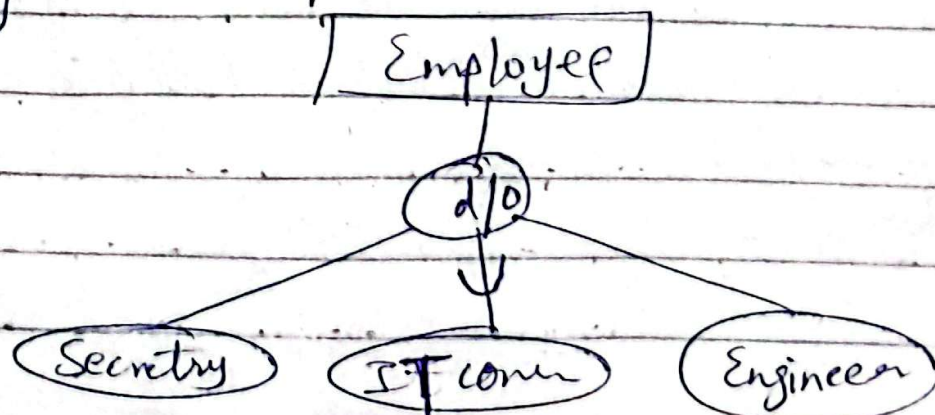
- ↳ Bottom to top process
- ↳ Start with sub classes
- ↳ Identify similarities.
- ↳ Attributes & relationship

- ① Disjoint (At a time only 1)
- ② Overlap (At a time more than 1)



- ↳ subclass considered as overlapping
- ↳ More than 1.

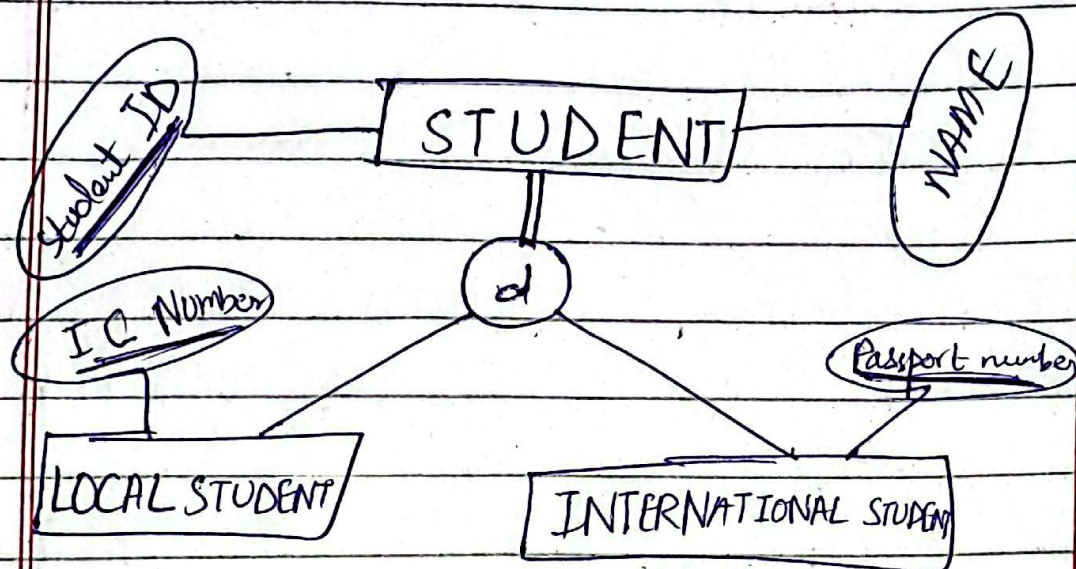
For disjoint ("d")
For overlap ("o")
e.g



→ Total specialisation

↳ Every entity must have a relationship with member

→ Partial specialisation



(must to "d"
compulsory)

Category

↳ subclass belong to one superclass.

↳ Category have subclasses & superclasses.

↳ When superclass is multiple then it is category.

→ In category superclasses have no relation b/w each other.

→ No direct relation of superclasses ~~by~~ but when inherited link with only one subclass.

③ things

① Local Attributes

② Relation

③ Attribute inheritance.

→ In category subclass member is also a member of superclass.

→ It inherit attributes of superclass's relationships

→ In category ("U") union